# Small Signal Modelling of PSFB-I

## Introduction

In the rapidly evolving landscape of electric vehicle charging and other high-power applications, Phase-Shifted Full-Bridge (PSFB) converters play a pivotal role due to their ability to achieve Zero Voltage Switching (ZVS) across a wide operating range, enhancing efficiency and reducing switching losses. However, the dynamic behavior of these converters, particularly under varying input voltages and load conditions, presents significant control challenges. To ensure stable and precise output voltage regulation, engineers must develop sophisticated control strategies. This lecture delves into the crucial technique of small-signal modeling for PSFB converters, providing the analytical framework necessary to understand how the converter responds to minor perturbations in its operating parameters. By dissecting the intricate relationships between duty cycle, input voltage, output current, and the resulting effective duty ratio, we lay the groundwork for designing robust feedback control systems that can maintain a constant output voltage despite external disturbances. This understanding is fundamental for optimizing the performance and reliability of modern charging infrastructure.

## Core Concepts

| Term | Definition | Significance |
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| `Phase-Shifted Full-Bridge (PSFB) Converter` | A DC-DC converter topology that uses phase-shifted PWM control to achieve Zero Voltage Switching (ZVS) for its primary switches, improving efficiency. | Essential for high-power, high-voltage applications like EV charging, where efficiency and reduced switching losses are critical. |
| `Zero Voltage Switching (ZVS)` | A switching technique where a power semiconductor device turns on when the voltage across it is zero, minimizing turn-on switching losses. | Crucial for high-frequency operation of PSFB converters, as it significantly reduces power dissipation and allows for smaller components. |
| `Leading Leg` | One half-bridge of the full-bridge converter where ZVS is typically easier to achieve due | Its ZVS capability is generally robust across most load conditions, |

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| | to the large output inductance. | contributing to overall converter efficiency. |
| `Lagging Leg` | The other half-bridge of the full-bridge converter where ZVS is more challenging to achieve, especially at light loads, due to reliance on leakage inductance. | Its ZVS performance dictates the converter's efficiency at light loads and requires careful design of dead time and parasitic elements. |
| `Leakage Inductance (L_leakage)` | The parasitic inductance inherent in the transformer windings, which plays a critical role in facilitating ZVS in PSFB converters. | Utilized in conjunction with switch output capacitances to create resonant conditions for ZVS, but also contributes to duty cycle loss. |
| `Duty Loss (D_loss)` | A reduction in the effective duty cycle of the converter, primarily caused by the ZVS transition intervals and the interaction of leakage inductance and output capacitances. | Directly impacts the output voltage regulation, as the actual output voltage depends on the effective duty cycle, not just the applied duty cycle. |
| `Effective Duty Ratio (D_effective)` | The actual duty cycle that contributes to power transfer, calculated as the applied duty ratio minus the duty loss (D - D_loss). | This is the true control variable that determines the output voltage, making its accurate modeling essential for precise control. |
| `N_IL` | The output inductor current reflected to the primary side of the transformer, a key current component influencing the ZVS transitions and duty loss. | Its magnitude affects the energy available for charging/discharging switch capacitances, directly impacting ZVS conditions and the extent of duty loss. |

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| `Small Signal Model` | A linearized representation of a non-linear system around an operating point, used to analyze its dynamic response to small perturbations. | Indispensable for designing stable and responsive feedback control loops for power converters, allowing for prediction of transient behavior. |

### Detailed Analysis

The lecture focuses on developing a small-signal model for the Phase-Shifted Full-Bridge (PSFB) converter, which is crucial for designing effective control systems. This involves understanding the converter's operating modes, the phenomenon of Zero Voltage Switching (ZVS), and how various parameters influence the effective duty ratio.

#### 1. PSFB Operating Modes and ZVS Conditions

The PSFB converter operates through six distinct modes, which repeat cyclically. These modes involve the sequential turning on and off of diagonal switches (e.g., S1 and S4, then S2 and S3) and the conduction of output rectifier diodes (DR1, DR2). A key feature of the PSFB is its ability to achieve ZVS for all primary switches, which significantly reduces switching losses.

- **ZVS Mechanism**: ZVS is achieved by utilizing the output capacitances of the MOSFETs and the transformer's leakage inductance ( `L_leakage` ). During switch transitions, these parasitic elements resonate, allowing the voltage across a switch to drop to zero before it turns on.
- **Leading Leg ZVS**: For the leading leg switches (e.g., S1, S2), ZVS is generally robust across a wide load range. This is because the large output inductance, reflected to the primary side ( `N_IL` ), provides sufficient current to charge and discharge the switch capacitances (C1, C2) to either the input voltage ( `Vin` ) or zero. Even with small `N_IL` , 100% ZVS is often achievable.
- **Lagging Leg ZVS**: ZVS for the lagging leg switches (e.g., S3, S4) is more challenging, especially at light load conditions. Here, the energy for charging/discharging capacitances (C3, C4) primarily comes from the current flowing through `L_leakage` . If the primary current ( `Ip` ) is very small (low load), the energy might be insufficient to fully discharge C3 and

charge C4, leading to a loss of ZVS. This condition is exacerbated by smaller `L_leakage` values, larger C3/C4, or higher input voltages.

- **Dead Time Requirement**: Sufficient dead time must be provided between the turn-off of one switch and the turn-on of its complementary switch in a leg. This dead time allows the body diode of the MOSFET to conduct, ensuring the voltage across the device is zero before the gate pulse is applied, thus achieving ZVS. The dead time for the leading leg is approximately `2C * Vin / I1`.

### 2. Duty Loss and Effective Duty Ratio

A critical consequence of achieving ZVS in PSFB converters is the phenomenon of "duty loss." During the ZVS transition intervals (specifically from mode 2 to mode 5 in the described operation), the converter experiences a period where the primary voltage is effectively zero, leading to a reduction in the effective duty cycle.

- **Origin of Duty Loss**: The duty loss occurs because the primary current ( `Ip` ) needs time to resonate with the switch capacitances and `L_leakage` to facilitate ZVS. During this resonance, the voltage across the primary side of the transformer ( `Vab` ) might be zero or changing, not contributing to power transfer in the same way as the main conduction intervals.
- **Duty Loss Expression**: The lecture provides the duty loss expression as:
  `D_loss = 4 * N * L_leakage * I_naught * F_SW / Vin`
  where `N` is the turns ratio, `L_leakage` is the leakage inductance, `I_naught` is the output current, `F_SW` is the switching frequency, and `Vin` is the input voltage.
- **Effective Duty Ratio**: The output voltage of a PSFB converter is analogous to a buck converter, given by `V0 = N * Vin * D_effective`. However, due to duty loss, the `D_effective` is not simply the applied duty ratio ( `D` ) but rather:
  `D_effective = D - D_loss`
  This means that the actual control variable for the output voltage is `D_effective`, which is influenced by `D`, `L_leakage`, `I_naught`, `F_SW`, and `Vin`.

### 3. Control Objectives and Buck-Derived Topology

The primary control objective for a PSFB converter is to regulate the output voltage ( `V0` ) to a desired reference value.

- **Buck-Derived Nature**: The output stage of the PSFB converter (comprising the output inductor `L`, capacitor `C`, and load `RL`) is identical to that of a

buck converter. This makes the PSFB a "buck-derived topology," and its output voltage relationship `V0 = N * Vin * D_effective` reflects this.

- **Challenges in Control**: The presence of `D_loss` complicates control. When the applied duty ratio ( `D` ) is changed to regulate `V0` , `I_naught` (output current) also changes. Since `D_loss` is dependent on `I_naught` , a change in `D` indirectly affects `D_loss` , which then further modifies `D_effective` . Similarly, changes in `Vin` also impact `D_loss` and thus `D_effective` . This creates a complex dynamic interaction that needs to be modeled for accurate control.

### 4. Impact of Inductor Current Change on Effective Duty Ratio ( `di` term)

When the output inductor current ( `IL` ) changes, it directly impacts the `D_loss` and, consequently, the `D_effective` . This interaction needs to be captured as a small-signal term, `di` .

- **Mechanism**: A change in `IL` (represented as `il_tilde` in small-signal analysis) leads to a change in the reflected current `N_IL` . This change alters the time required for the primary current ( `Ip` ) to reach `N_IL` during the ZVS transition, which in turn changes the `delta T` (the time duration contributing to duty loss).
- **Derivation of `delta T`** : The lecture explains that `delta T` is related to the slope of the primary current during the ZVS transition, which is `Vin / L_leakage` . A change in `IL` results in a `delta T` given by:
  `delta T = L_leakage * 2 * N * il_tilde / Vin`
- **Derivation of `di`** : The small-signal change in the effective duty ratio due to `il_tilde` is `di = - delta T / (Ts/2)` . The negative sign is crucial because if `IL` increases, `delta T` increases, meaning more time is spent in the ZVS transition, which reduces the effective duty ratio.
  Substituting `delta T` :
  `di = - (L_leakage * 2 * N * il_tilde / Vin) / (Ts/2)`
  Since `Ts = 1/F_SW` , this simplifies to:
  `di = - 4 * N * L_leakage * F_SW * il_tilde / Vin`
  This `di` term represents an additional dynamic component that modifies `D_effective` whenever `IL` changes.

### 5. Impact of Input Voltage Change on Effective Duty Ratio ( `dv` term)

Similarly, a change in the input voltage ( `Vin` ) also affects `D_loss` and `D_effective` , requiring a small-signal term, `dv` .

- **Mechanism**: A change in `Vin` (represented as `vin_tilde` ) alters the slope of the primary current ( `Vin / L_leakage` ) during the ZVS transition. If `Vin` increases, the slope increases, causing the current to reach `N_IL` faster, thereby reducing the `delta T` associated with duty loss.
- **Derivation of `delta T`** : The lecture derives `delta T` due to `vin_tilde` as:
  `delta T = N * (2 * I0 - V0/L * (1-D) * Ts/2) * L_leakage * vin_tilde / Vin^2`
  Here, `I0` is the average output current, `V0` is the output voltage, `L` is the output inductance, `D` is the steady-state duty ratio, and `Ts` is the switching period. The term `(2 * I0 - V0/L * (1-D) * Ts/2)` represents the current swing.
- **Derivation of `dv`** : The small-signal change in the effective duty ratio due to `vin_tilde` is `dv = delta T / (Ts/2)` . The positive sign is used here because if `Vin` increases, `delta T` reduces, which means `D_loss` reduces, and thus `D_effective` increases. This is a positive effect on `D_effective` .
  Substituting `delta T` :
  `dv = (N * (2 * I0 - V0/L * (1-D) * Ts/2) * L_leakage * vin_tilde / Vin^2) / (Ts/2)`
  This `dv` term captures the additional dynamic modification to `D_effective` when `Vin` changes.

### 6. Combined Dynamics of Effective Duty Ratio

The overall change in the effective duty ratio ( `d_effective_tilde` ) is a summation of the actual small change in the applied duty ratio ( `d_tilde` ) and the additional dynamic terms `di` and `dv` :
`d_effective_tilde = d_tilde + di + dv`
This comprehensive understanding of `D_effective` 's dynamics is essential for accurately modeling the PSFB converter's response to control inputs and disturbances. The small-signal model will integrate these terms to predict how the output voltage responds to changes in `D` , `IL` , and `Vin` .

## Key Takeaways

- [ ] PSFB converters utilize phase-shifted PWM to achieve Zero Voltage Switching (ZVS), improving efficiency.
- [ ] ZVS for the leading leg is generally robust, while ZVS for the lagging leg is challenging at light loads.
- [ ] Transformer leakage inductance ( `L_leakage` ) and switch output capacitances are critical for ZVS.
- [ ] ZVS operation introduces "duty loss" ( `D_loss` ), which reduces the effective duty ratio.
- [ ] The output voltage is determined by the effective duty ratio: `V0 = N * Vin * (D - D_loss)` .
- [ ] Changes in output inductor current ( `IL` ) cause an additional change in `D_effective` (the `di` term), which is inversely proportional to `IL` .
- [ ] Changes in input voltage ( `Vin` ) also cause an additional change in `D_effective` (the `dv` term), which is directly proportional to `Vin` .
- [ ] The total effective duty ratio is a combination of the applied duty ratio and these `di` and `dv` dynamic terms.
- [ ] Small-signal modeling is necessary to account for these complex interactions and design stable control loops.

## Conclusion

The detailed analysis of small-signal dynamics in PSFB converters, particularly the derivation of the `di` and `dv` terms, highlights the intricate relationship between the applied duty ratio, input voltage, and output current in determining the actual effective duty ratio. This understanding moves beyond a simplistic view of converter operation, acknowledging the inherent duty loss and its variability under different operating conditions. By meticulously modeling how changes in inductor current and input voltage propagate through the system to affect the effective duty cycle, we gain the necessary insights to predict the converter's transient behavior. This foundational work in small-signal modeling is not merely an academic exercise; it is an indispensable step in designing robust and high-performance feedback control systems for PSFB converters. Such controllers are vital for maintaining stable output voltage in demanding applications like electric vehicle charging, where input voltage fluctuations and dynamic load changes are common. The next steps will involve integrating these derived small-signal terms into a complete transfer function, enabling the design of precise and resilient control strategies.